

SEMINAR PRESENTATION

Integrated Agrivoltaic–Anaerobic Digestion Optimization Across Four Climate Zones

*A Novel Extended Land Equivalent Ratio Framework
for Food-Energy-Water Nexus Assessment*

Sites: Konya (Türkiye) · Almería (Spain) · Ouagadougou (Burkina Faso) · Freiburg (Germany)

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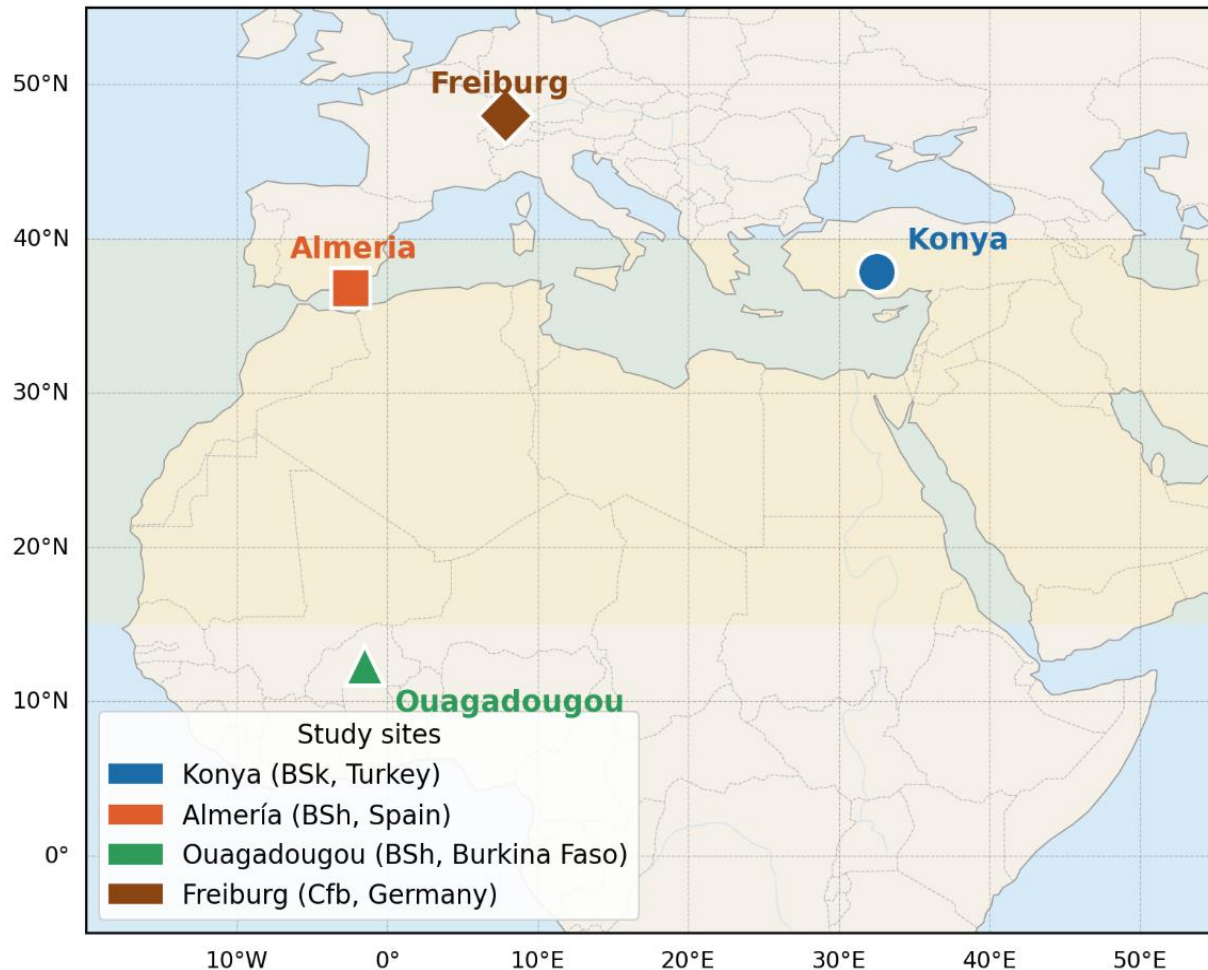
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Methodology

Framework Design, Models & Optimization

Study Sites

- Konya (Turkey)
- Almería (Spain)
- Ouagadougou (Burkina Faso)
- Freiburg (Germany)



PV Model & Crop Yield Equations

Cell Temperature — Faïman (2008) (Eq. 9)

$$T_{cell(t)} = T_{amb(t)} + \frac{G_{eff(t)}}{(U_0 + U_1 \times u(t))}$$

$U_0 = 25 \text{ W/m}^2\cdot\text{K}$ | $U_1 = 6.84 \text{ W}\cdot\text{s/m}^3\cdot\text{K}$ (wind-dependent convective cooling)

Crop Yield Ratio — PAR Integral (Eq. 11)

$$\frac{Y_{crop}}{Y_{ref}} = \frac{\int_{t_0}^{t_1} \min(PAR_{AV}(t), PAR_{sat}) dt}{\int_{t_0}^{t_1} PAR_{open}(t) dt}$$

PAR_{sat} (tomato) = 174 W/m^2 vs. lettuce = 76 W/m^2

PV Power Output (Eq. 10)

$$P_{PV(t)} = P_{STC} \times N \times (1 - \eta_{loss}) \times [1 + \beta_p(T_{cell(t)} - 25)] \times \frac{G_{eff(t)}}{1000}$$

$P_{STC} = 550 \text{ Wp}$ (Jinko Tiger Neo bifacial) $\beta_p = -0.0035 \text{ K}^{-1}$ $\eta_{loss} = 0.12$

Anaerobic Digestion & Arrhenius Correction

Arrhenius BMP Temperature Correction (Eq. 2)

$$BMP_T = BMP_{ref} \times \exp(\theta \times (T - T_{ref}))$$

$$\theta = 0.069 \text{ } ^\circ\text{C}^{-1} \quad T_{ref} = 37^\circ\text{C}$$

Digester Heat Balance (Eq. 18)

$$Q_{heat} = Q_{loss,cond} + Q_{inflow} - Q_{reaction}$$

$$Q_{loss} = U_{eff} \times A_{wall} \times (T_{dig} - T_{amb}) \times \frac{24}{1000}$$

$$PV \text{ supplies: } f_{PV,heat} \times E_{PV} (\eta_{elec} = 0.95)$$

Energy Self-sufficiency Ratio (ESR)

$$ESR = \frac{(E_{PV} + E_{biogas})}{Q_{heat}} \gg 1$$

The eLER Metric — Central Contribution

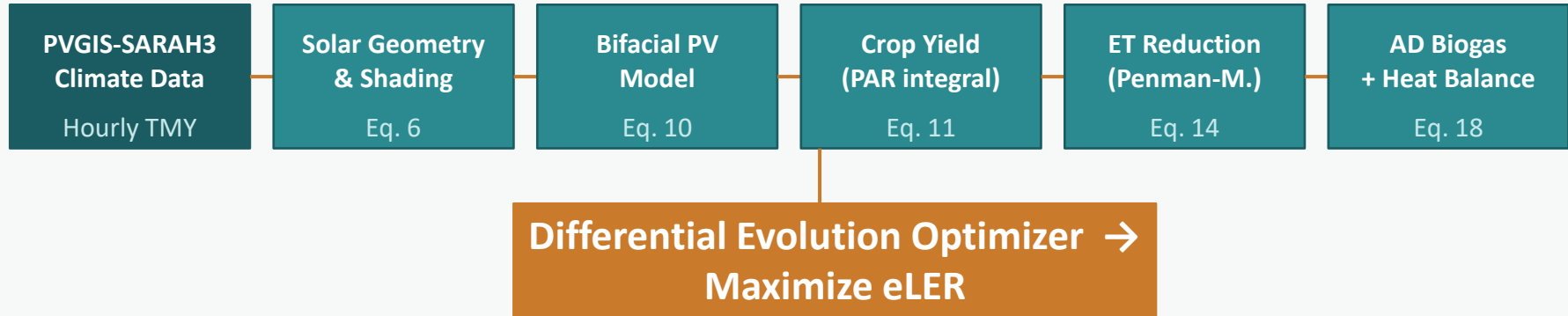
The eLER extends the classical two-component LER to include net biogas energy as a third land-use output:

$$eLER = \frac{Y_{crop,AV}}{Y_{crop,open}} + \frac{E_{PV,net}}{E_{PV,ref}} + \frac{E_{biogas,net}}{E_{biogas,ref}}$$

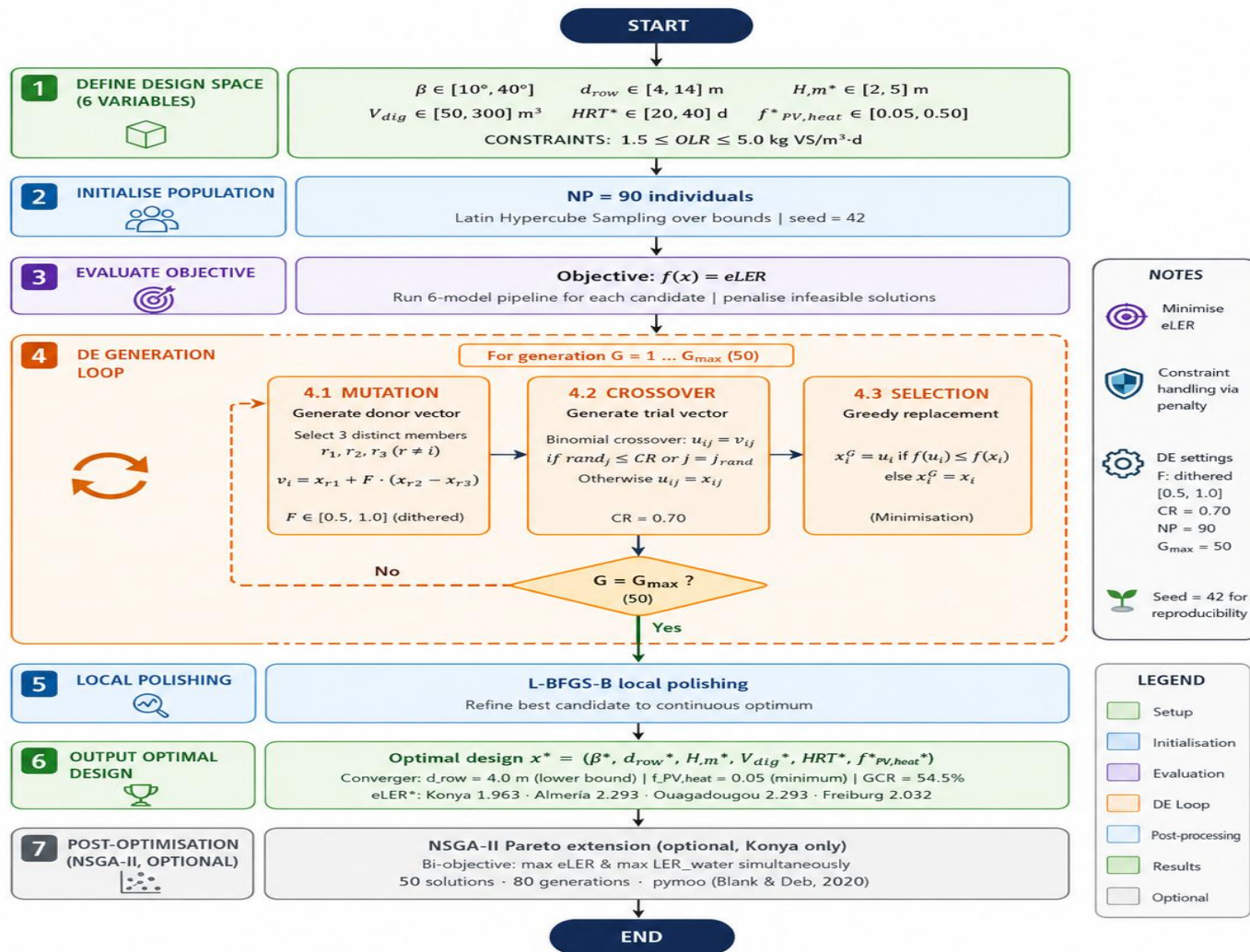
Study	Location	Crop	LER	System type
Dupraz et al. (2011)	France	Durum wheat	1.35	Fixed-tilt, modelled
Mohammedi et al. (2023)	Italy	Tomato	1.42	Fixed-tilt, exp.
Riaz et al. (2022)	USA (AZ)	Tomato	1.94 ★	Bifacial, optimized
Mazzeo et al. (2025)	Italy	Lettuce	1.85	Open-field, optimized

Integrated Modelling Framework

Six coupled sub-models simultaneously optimized over six design variables:



Optimization Design & AD Scenarios : Differential Evolution Optimizer



Optimization Design & AD Scenarios

Table 7: AD Valorization Scenarios (S0–S5)

Scenario	Description	Key change
S0	Baseline — mono-digestion	BMP_mult = 1.00
S1	Co-digestion (+30% manure)	+15% BMP
S2	Sub-optimal CH ₄ capture	$\eta_{\text{cap}} = 0.85$
S3	LPG displacement (on-site)	p_biogas = \$0.18/kWh
S4	Digestate N-P-K credit	N credit \$1.20/kg
S5	Biomethane grid injection ★	5.5× electricity price

★ S5 motivated by Nik Zad et al. (2026): biomethane = 5.5× electricity market price (Italian / EEG 2023)

* Open-field German yield (DLG, 2023); residue input = 24 t/ha vs. 48 t/ha at semi-arid sites — propagated through all economic & eLER calculations

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Data Inputs

PVGIS-SARAH3 Climate Data · Seasonal Profiles

PVGIS TYPICAL METEROLOGICAL YEAR (TMY) – SARA3

- Date & time (UTC)
- T2m [°C] - Dry bulb (air) temperature.
- RH [%] - Relative Humidity.
- G(h) [W/m²] - Global horizontal irradiance.
- Gb(n) [W/m²] - Direct (beam) irradiance.
- Gd(h) [W/m²] - Diffuse horizontal irradiance
- IR(h) [W/m²] - Infrared radiation downwards.
- WS10m [m/s] - Windspeed.
- WD10m [°] - Wind direction.
- SP [Pa] - Surface (air) pressure.

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PHOTOVOLTAIC GEOGRAPHICAL INFORMATION SYSTEM

European Commission

European Commission > PVGIS > Tools > Interactive tools

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Cursor: Selected: Select location! Elevation (m): Use terrain shadows: ☒ Calculated horizon ☐ Upload horizon file [Download csv](#) [Browse...](#) No file selected.

GRID CONNECTED

TRACKING PV OFF-GRID MONTHLY DATA DAILY DATA HOURLY DATA TMY

PERFORMANCE OF GRID-CONNECTED PV

Solar radiation database* Crystalline silicon

PV technology* Crystalline silicon

Installed peak PV power [kWp] 1

System loss [%] 14

Fixed mounting options

Mounting position* Free-standing

Slope [°] 35 ☐ Optimize slope

Azimuth [°] 0 ☐ Optimize slope and azimuth

☐ **PV electricity price**

PV system cost (your currency)

Interest [%/year]

Lifetime [years]

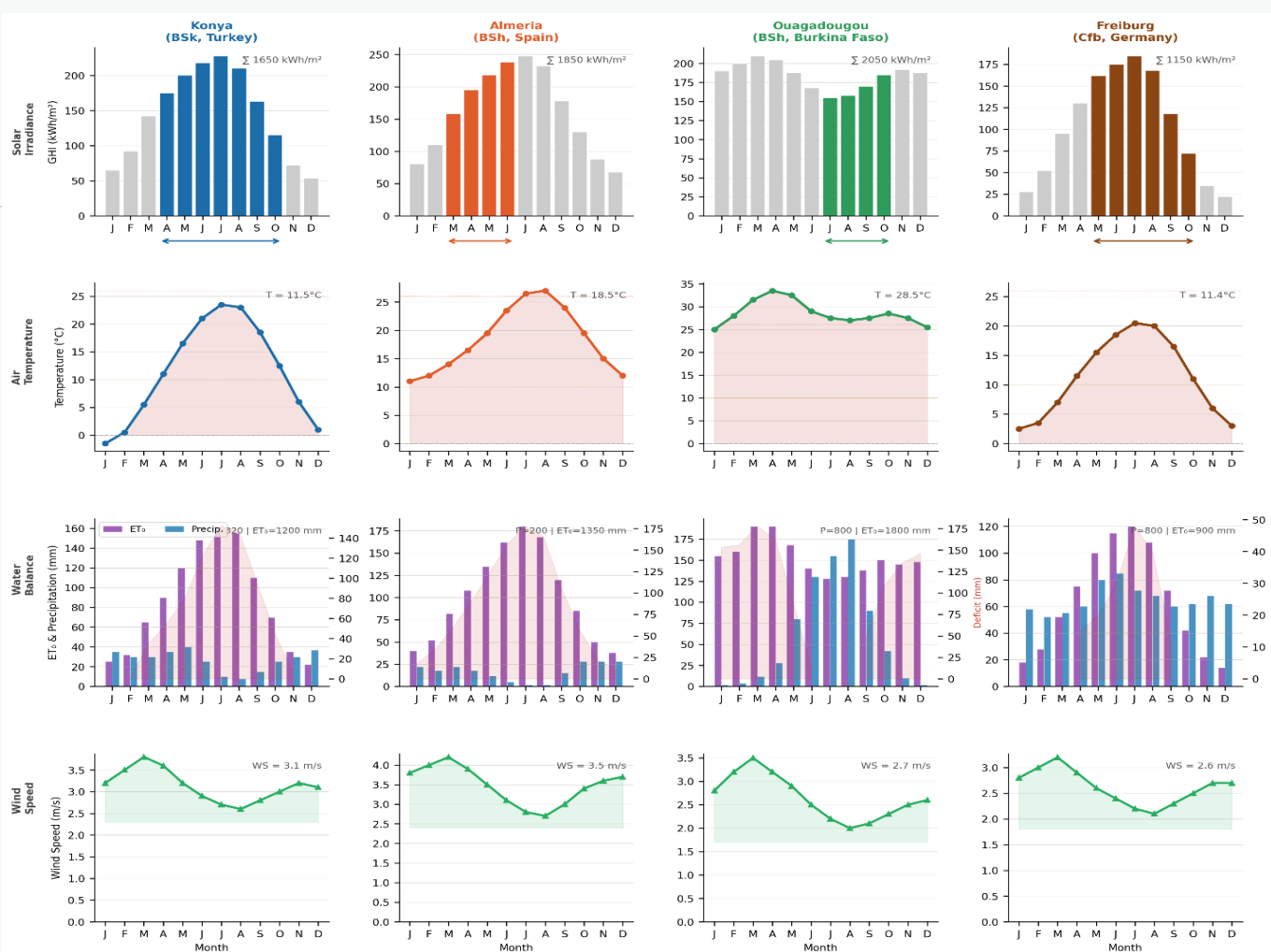
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Climate Data & Seasonal Profiles

PVGIS-SARAH3 Typical

Meteorological Year (TMY) hourly
data — mean bias error < $\pm 5\%$ for
monthly GHI (Huld et al., 2012;
Urraca et al., 2017). DNI and DHI
extracted directly (not
decomposed from GHI).



Study Sites & Climate Zones

Table 4: Geographical & climatic characteristics (PVGIS-SARAH3 TMY)

Parameter	Konya (BSk)	Almería (BSh)	Ouagadougou (BSh)	Freiburg (Cfb)
Latitude	37.87°N	36.83°N	12.37°N	47.99°N
Köppen climate	BSk cold semi-arid	BSh hot semi-arid	BSh tropical semi-arid	Cfb oceanic temp.
Mean annual GHI (kWh/m²)	1,650	1,850	2,050	~1,150
Mean annual T (°C)	11.5	18.5	28.5	~11.4
Summer Tmax (°C)	38	35	41	~35
Winter Tmin (°C)	-10	+5	+15	~-12
Annual ET _o (mm)	1,200	1,350	1,800	~900
Annual Precip. (mm)	320	200	800	~800

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Results

Optimization · eLER · Energy · Water · Scenarios

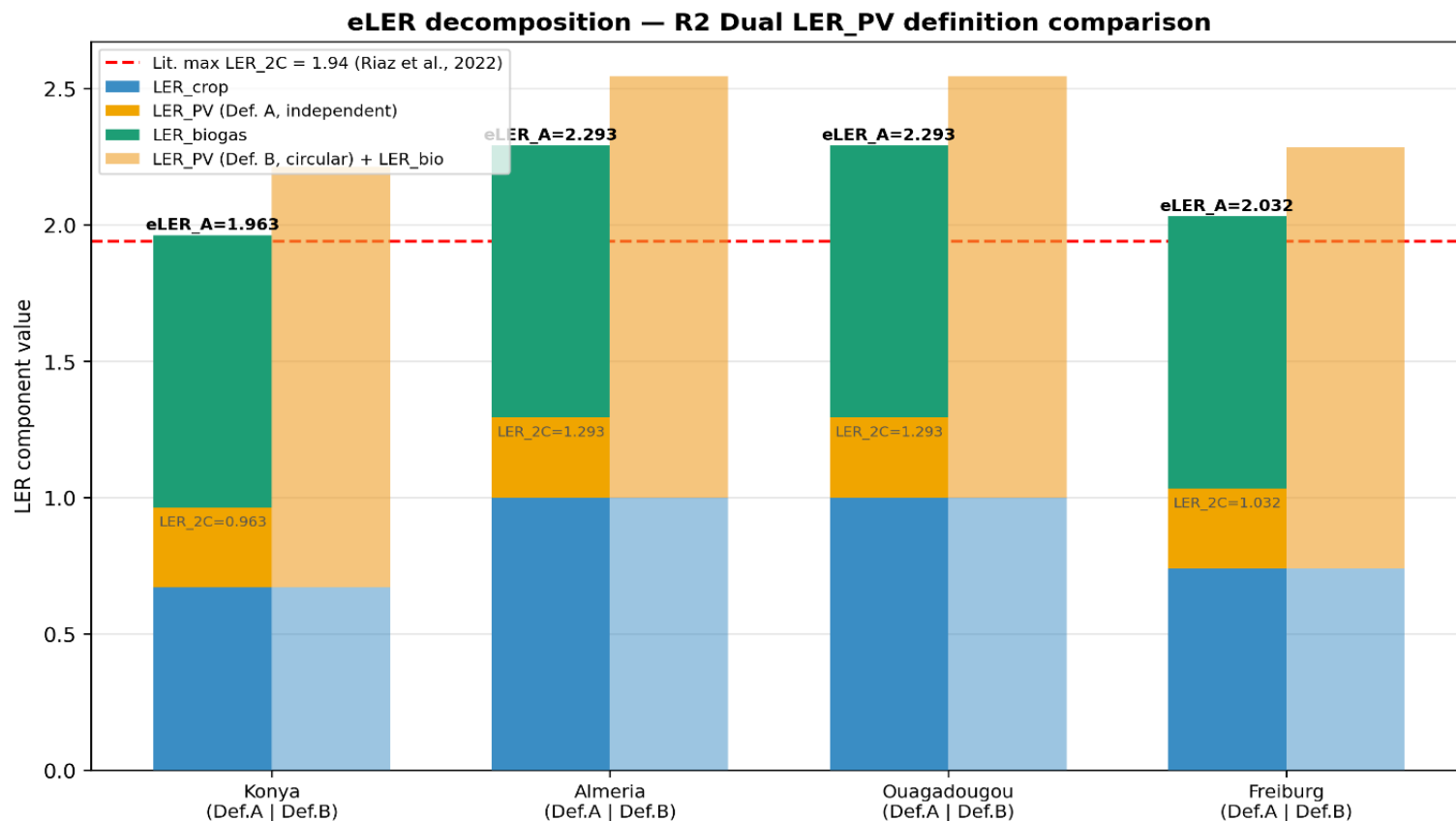
Optimal Design Parameters

Table 9: Optimal design variables and primary performance metrics — all four sites

Parameter	Konya (BSk)	Almería (BSh)	Ouagadougou (BSh)	Freiburg (Cfb)
Tilt angle β (°)	33.3	19.3	26.0	15.9
Row spacing d_{row} (m)	4.00	4.00	4.00	4.00
Module height H_m (m)	2.02	3.03	2.48	2.69
Digester volume V_{dig} (m³)	64	80	114	53
HRT design (days)	29.1	21.8	22.8	20.8
PV heating fraction $f_{PV,heat}$	0.050	0.050	0.050	0.050
GCR (%)	54.5	54.5	54.5	54.5
OLR (kg VS/m³·d)	4.35	4.67	3.13	3.70

Key finding: Optimizer converges to $d_{row} = 4.0$ m (lower bound) and $f_{PV,heat} = 0.05$ (minimum) at ALL four sites → GCR = 54.5% everywhere. Boundary convergence indicates eLER-maximizing solution lies at the edge of the feasible domain.

eLER Decomposition — Main Result



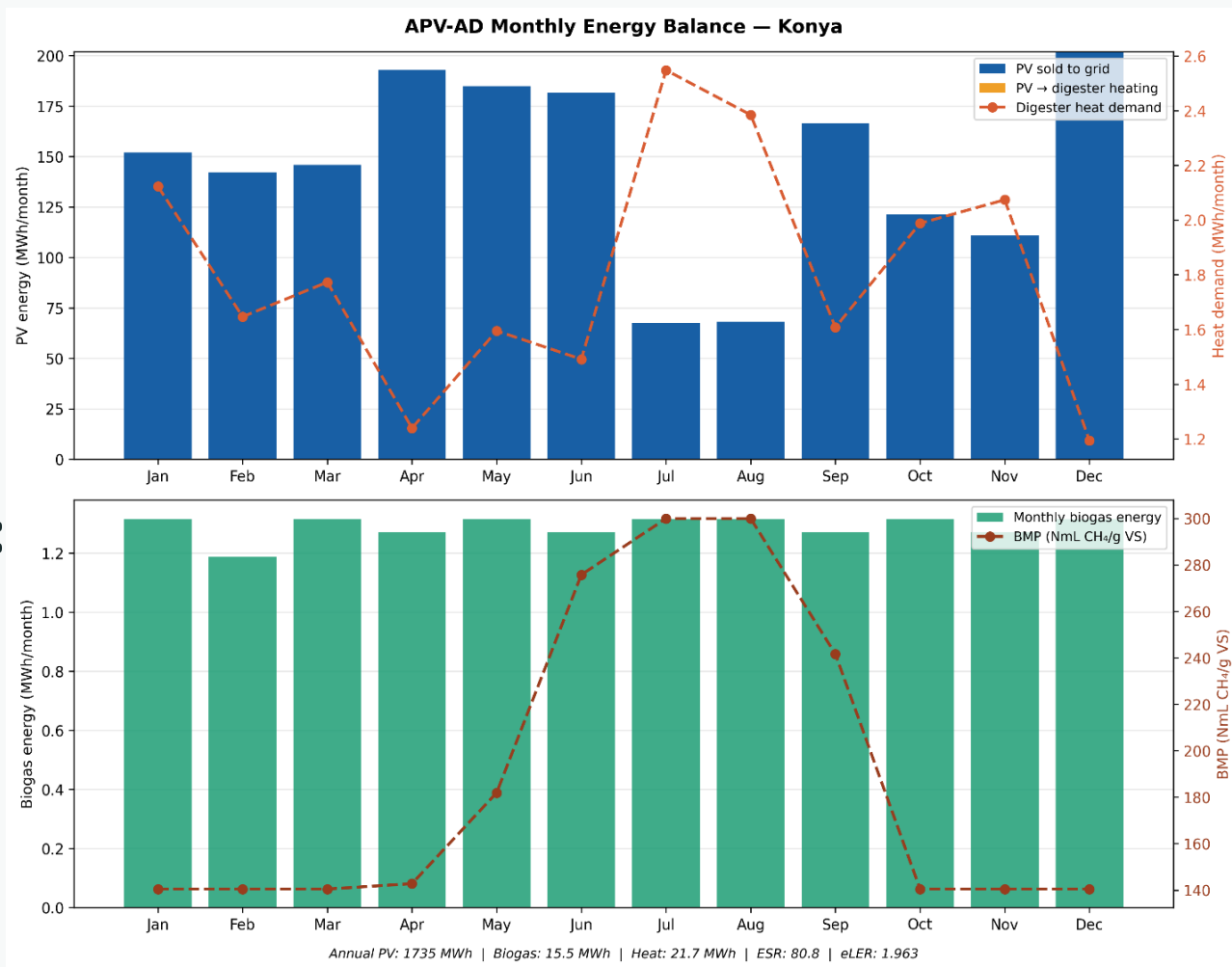
eLER Decomposition — Main Result

Table 10: eLER decomposition, all four sites (Scenario S0, Baseline)

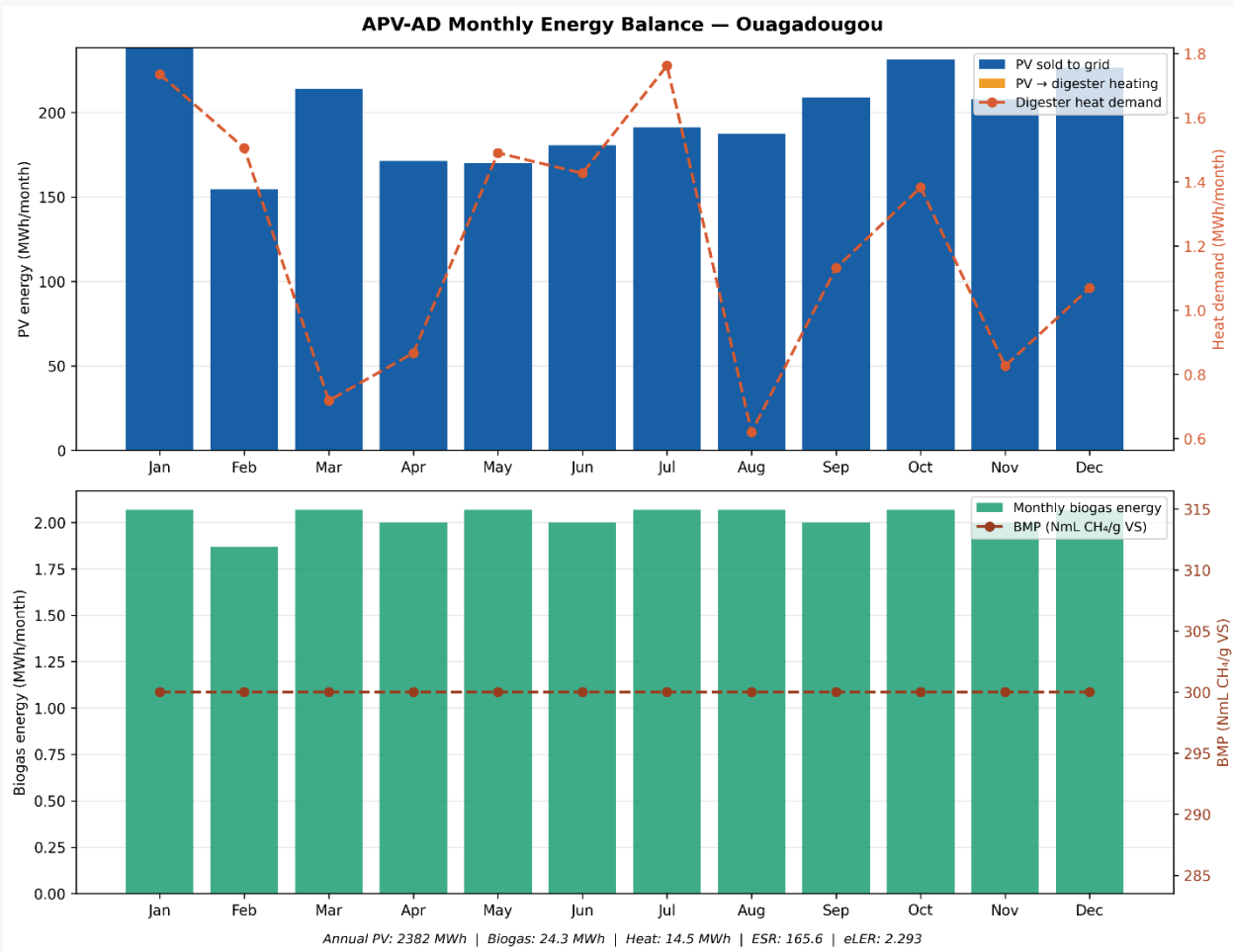
Metric	Konya (BSk)	Almería (BSh)	Ouagadougou (BSh)	Freiburg (Cfb)
LER_crop	0.670	1.000 ✦	1.000 ✦	0.739
LER_PV (Def. A — independent)	0.293	0.293	0.293	0.293
LER_PV (Def. B — circular = GCR)	0.545	0.545	0.545	0.545
LER_biogas	1.000	1.000	1.000	1.000
eLER_global (Def. A)	1.963	2.293	2.293	2.032
LER_2C = LER_crop + LER_PV(A)	0.963	1.293	1.293	1.032
Lit. max LER_2C (Riaz et al. 2022)	1.94 ★	1.94 ★	1.94 ★	1.94 ★

✦ Penalty-free PV: LER_crop = 1.000 at Almería & Ouagadougou — high solar elevation → near-zero shading during growing season

Annual Energy Performance & Thermal Balance : Konya



Annual Energy Performance & Thermal Balance : Ouagadougou



Annual Energy Performance & Thermal Balance

Table 11: Annual energy performance — all four sites

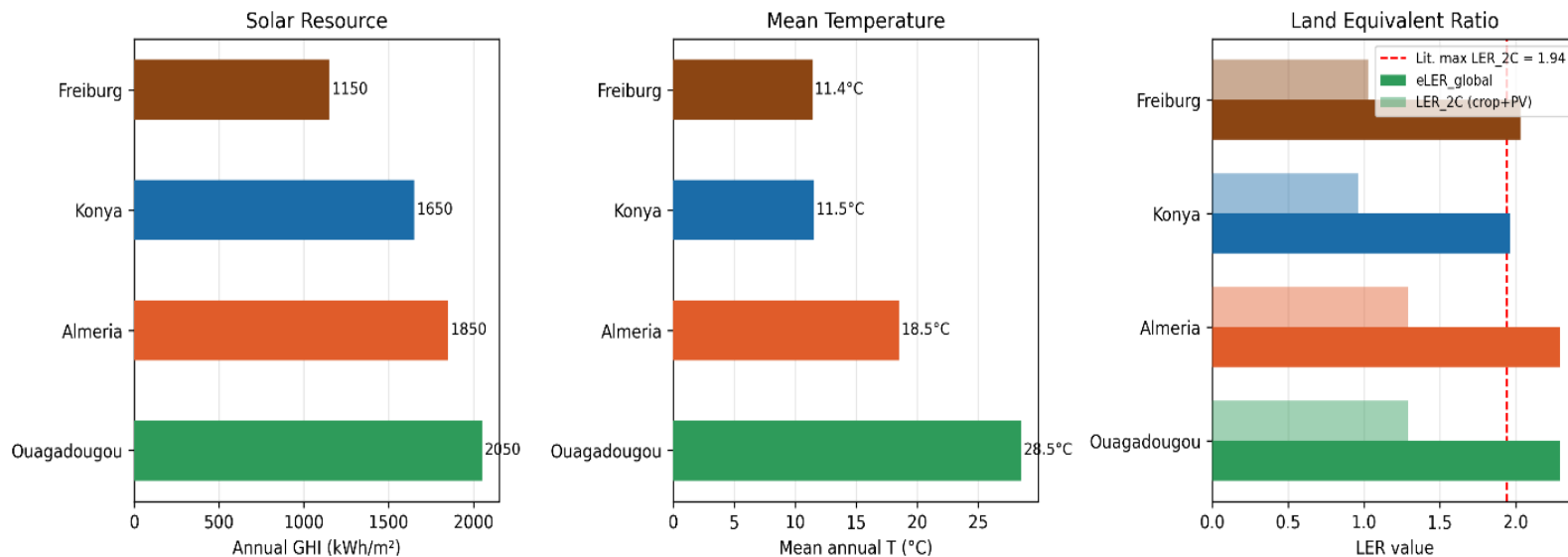
Parameter	Konya	Almería	Ouagadougou	Freiburg
PV total (MWh/ha·yr)	1,735	2,227	2,382	1,560
PV sold to grid (MWh/ha·yr)	1,649	2,116	2,263	1,482
Annual biogas total (MWh/ha)	15.5	17.8	24.3	6.0
Annual heat demand (MWh/ha)	21.7	23.3	14.5	20.5
Biogas used for heating	0.0	0.0	0.0	0.0
ESR	80.8	96.2	165.6	76.3
BMP annual avg (NmL CH ₄ /gVS)	190.7	218.8	300.0	148.8

Climate Gradient: Counter-Intuitive eLER Ordering

GHI gradient: Ouagadougou > Almería > Konya > Freiburg

eLER gradient: Almería = Ouagadougou > Freiburg > Konya ← does NOT follow GHI !

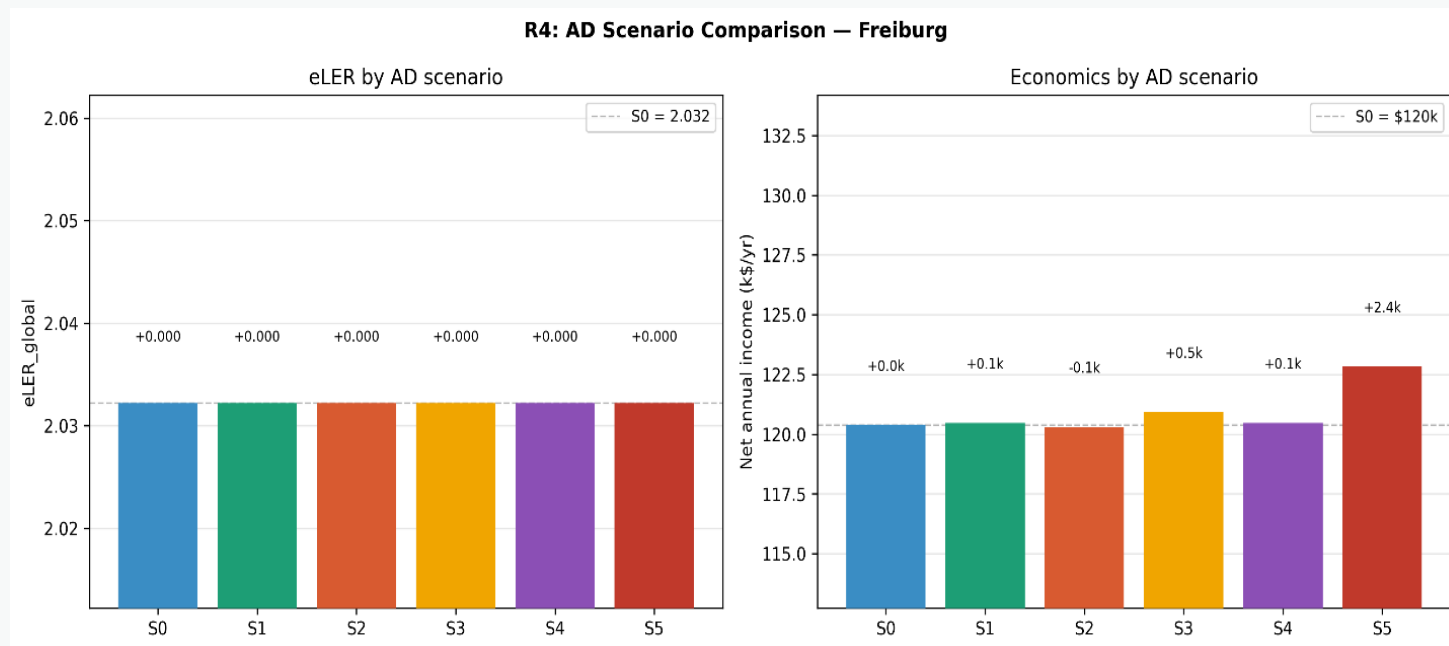
APV-AD Four-Site Climate Gradient
(Ouagadougou → Almería → Konya → Freiburg, decreasing GHI)



AD Scenario Economics (S0–S5)

eLER is invariant across all scenarios ($\Delta eLER = 0.000$ for S0–S5) — structural result. Economic differentiation only.

Net annual income by scenario — summary (k\$/yr):



AD Scenario Economics (S0–S5)

eLER is invariant across all scenarios ($\Delta eLER = 0.000$ for S0–S5) — structural result. Economic differentiation only.

Net annual income by scenario — summary (k\$/yr):

Scenario	Konya	Almería	Ouagadougou	Freiburg
S0 Baseline	\$91k	\$195k	\$267k	\$120k
S1 Co-digestion	\$91k	\$196k	\$267k	\$120k
S3 LPG displacement	—	—	+\$2.4k	—
S5 Biomethane ★	+\$3.9k	+\$7.2k	+\$14.1k	+\$2.4k
Payback period (yr)	9.4 yr	4.5 yr	3.3 yr	7.1 yr

★ S5 increments over S0 Baseline. S5 adds \$14.1k (+5.3%) at Ouagadougou (highest biogas volume). Freiburg S5 limited by 6.0 MWh/yr biogas. Note: figures are revenue minus OpEx — not full NPV/IRR.

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Limitations & Future Research

Scope, Uncertainty, and Open Questions

Limitations

L1 Crop yield model uncertainty

L2 ET model scope

L3 Digestate–soil feedback (Gap G6)

L4 Biomethane S5 — no upgrading CAPEX

L5 Economic analysis — no full NPV/IRR

L6 Single-crop analysis

Suggestions for Future Research

F1 Multi-crop validation

F2 eLER standardization — alternative LER_biogas

F3 Full NPV/IRR + LCOE with CAPEX breakdown

F4 Field validation — Freiburg / Baden-Württemberg

F5 Digestate-to-soil nutrient feedback (Gap G6)

F6 eLER confidence intervals

Key Conclusions

- 1 eLER exceeds the published 1.94 benchmark at ALL four climate zones (1.963–2.293).**
- 2 eLER is non-monotone with solar resource**
- 3 Penalty-free PV confirmed at Mediterranean & tropical sites**
- 4 Thermal self-sufficiency confirmed at all sites (ESR = 76–166).**
- 5 Biomethane S5 is the dominant economic scenario everywhere (+\$2.4–14.1k/yr).**

Thank You !

